A RECURSIVE BUCHSTAB SWITCHING IMPROVEMENT FOR THE GREATEST PRIME FACTOR OF $n^2 + 1$

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Abstract. Let $P^+(m)$ denote the greatest prime factor of m. Grimmelt and Merikoski recently proved unconditionally the Type I and Type II ranges that had previously yielded, under Selberg's eigenvalue conjecture, the exponent 1.312 for $P^+(n^2+1)$. Li subsequently announced the exponent 1.317 by sharpening the numerical Buchstab integrations. We use the same unconditional arithmetic input but recursively expose the one-prime tail in the linear-sieve range $7/6 < \alpha < 5/4$. The resulting three-prime children are bounded using either the available Type II estimate or a lower dimension-one linear sieve. An outward-rounded integer certificate, independently recomputed in Lean, fixed-width C++, and arbitrary-precision Python, proves that this switching saves at least 0.032303187971 on the normalized sieve scale. Together with the original published upper bounds and an independent elementary lower bound for the subtracted one-prime term, the same arithmetic yields

$$P^+(n^2 + 1) > n^{1.323}$$

for infinitely many positive integers n, once the cited Grimmelt–Merikoski Type I/II estimates, the standard dimension-one linear sieve, and the finite Mellin–Perron transfer are granted. The internal block exponent is 1.3231 and the exact final normalized bound is $8997005488261/9000000000000 < 1$. Lean 4 verifies the finite Buchstab sign ledger, the recursive integer switching certificate, the primitive outward-rounding rules, the convex ten-panel rational midpoint certificate for the elementary F_6 bound, the exact rational endpoint budget, and the final dyadic-to-pointwise real-power inequality. It does not verify the localization or prefix-uniform sieve transfer, nor the full analytic theorem. This is a quantitative weak form of Landau's fourth problem: it produces a prime divisor exceeding $n^{1.323}$ and hence a complementary cofactor smaller than $n^{0.677+o(1)}$, but it does not overcome the sieve parity barrier.

1. Introduction

Landau's fourth problem asks whether $n^2 + 1$ is prime for infinitely many positive integers n. Iwaniec proved that $n^2 + 1$ is a prime or a product of two primes for infinitely many n [4]; the possibility that all surviving values are semiprimes is the classical parity obstruction. A standard quantitative complement is to make $P^+(n^2 + 1)$ as large as possible. Merikoski proved that $P^+(n^2 + 1) > n^{1.279}$ infinitely often and obtained 1.312 conditionally on Selberg's eigenvalue conjecture [6]. Pascadi obtained the unconditional exponent 1.3 [7]. Grimmelt and Merikoski then recovered 1.312 unconditionally by proving uniform Type I and Type II estimates

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for roots of quadratic congruences [2]. Li announced 1.317, using sharper numerical bounds for the same Buchstab decomposition [5].

These exponents quantify progress toward Landau's problem: a bound $P^+(n^2+1) > n^\vartheta$ leaves a cofactor $m < n^{2-\vartheta+o(1)}$. Theorem 1.1 lowers this exponent to 0.677, while still leaving the parity distinction $m = 1$ versus $m > 1$ unresolved.

Our new input is a recursive Buchstab switching in the range where the old argument used the linear sieve. Merikoski already observed that Buchstab's identity could be applied more times there to generate additional Type II sums [6, Remark 12]. With the stronger range now available unconditionally, this produces a numerically substantial saving. In addition to retaining Type II three-prime children, we use the lower linear sieve for the complementary children and recursively upper-bound the two-prime descendants of the previously discarded one-prime tail. Li explicitly keeps the same sieve decomposition and changes only the numerical estimates for its terms [5, Lemmas 2.1 and 2.2]; in particular, the contribution on $7/6 < \alpha < 5/4$ remains the old linear-sieve term minus the one-prime Type II term. The three-prime term below is therefore absent from the previously announced 1.317 argument.

Theorem 1.1. *There are infinitely many positive integers n such that*

$$P^+(n^2+1) > n^{1323/1000} = n^{1.323}.$$

Remark 1.2 (Proof-status boundary). The theorem is presented as the paper's candidate result conditional on the cited Grimmelt–Merikoski Type I/II estimates, the standard dimension-one linear sieve, and the finite Mellin–Perron transfer in Section 3. The source transfer in Section 2 records the cited external inputs and their normalization. These conventional analytic inputs are not formalized in Lean. Lean verifies the finite Buchstab signs, integer certificate, outward-rounding rules, proof-adverse cell geometry, model-normalization algebra, the elementary F_6 midpoint/convexity certificate, the exact rational endpoint budget, and the final real-power conversion. Small audit lemmas additionally check only affine interior margins, the integer half-threshold, and an abstract representation-count bound; they do not prove the Perron or prefix-sieve transfer. The formalization should not be read as having verified the full analytic theorem or solved Landau's fourth problem.

Remark 1.3 (Comparison with an isolated larger claim). Carella's arXiv note [1, Section 5, (5.3)–(5.4)] claims the exponent 3/2, but the transition from its (5.3) to (5.4) replaces

$$\sum_{d|m} \Lambda(d) = \log m \quad \text{by} \quad \sum_{p|m} \log p = \log \operatorname{rad}(m)$$

while retaining a lower bound. The latter quantity is at most the former, and n^2+1 need not be squarefree (for example, $7^2+1 = 2\cdot 5^2$). That step therefore does not establish the claimed bound. Our comparison frontier is the Grimmelt–Merikoski theorem and Li's subsequent 1.317 calculation.

The gain over 1.317 is 0.006. The argument is computer assisted only in the evaluation of an explicit finite family of piecewise-rational integrals. The certificate uses no floating-point operations. We retain Merikoski's original, relatively loose upper bounds for all positive Buchstab deficiencies before 7/6, so Li's Mathematica integrations are not an input.

2. Transferred arithmetic information

We use the notation of [6, Section 2]. Let b be a fixed nonnegative smooth function supported on $[x, 2x]$, put

$$|\mathcal{A}_d| = \sum_{\ell^2+1\equiv 0 \pmod d} b(\ell), \qquad X_b = \int_{\mathbb{R}} b(t)\,dt,$$

and let

$$\rho(d) = \#\{\nu \pmod d : \nu^2 + 1 \equiv 0 \pmod d\}.$$

The model sequence replaces $|\mathcal{A}_d|$ by $X_b\rho(d)/d$. For a smooth modulus weight ψ_P supported on $[P, 4P]$, write

$$S(x, P) = \sum_p \psi_P(p)|\mathcal{A}_p| \log p.$$

Proposition 2.1 (Grimmelt–Merikoski transfer)**.** *Fix a sufficiently small interior margin $\eta > 0$. The Type I estimate used in* [6, Proposition 3] *is available with $D \le x^{1/2-\eta}$. If $MN = x^\alpha$ and $M \ge N$, the Type II estimate in* [6, Proposition 4(i)] *is available, with a power-saving error, throughout every closed interior subrange of*

$$x^{\alpha-1} < N < x^{(2-\alpha)/3}, \tag{1}$$

for divisor-bounded coefficients whose N-coefficient is supported on squarefree integers.

Proof. Take the fixed shift $a = h = 1$ in [2, Corollaries 7.1 and 7.2]. Corollary 7.1 gives the Type I level $x^{1/2-\eta}$. More precisely, it bounds the sum over $d \le D$ of the absolute value of each modulus-d discrepancy. Therefore the triangle inequality permits arbitrary divisor-bounded coefficients λ_d, at a cost $x^{o(1)}$ absorbed by the displayed power saving. Corollary 7.2 gives

$$x^{\alpha-1+2\eta} \le N \le x^{(2-\alpha)/3-4\eta/3}.$$

Since η is arbitrary, this supplies every fixed closed interior range in (1). Its coefficient hypotheses are exactly those stated above. Also $N \le x^{1/3}$, so $M = x^\alpha/N \ge N$ in the ranges used here. The smooth-weight and Mellin separations are those already used in [6, Section 3.4].

More explicitly, the error exponents in Corollaries 7.1 and 7.2 are, up to an $x^{o(1)}$ factor,

$$1 - (1 - 2\theta)\eta + \varepsilon/4, \qquad 1 - (1 - 2\theta)\eta + \varepsilon/8.$$

Unconditionally $\theta = 7/64$. For example, after fixing the sieve margin, choosing $\varepsilon = \eta$ gives respectively $1-17\eta/32$ and $1-21\eta/32$; the logarithmic losses below are therefore absorbed by a fixed power saving. For $a = h = 1$, the auxiliary character condition in [2, Theorem 1.1] is unconditional by the zero-free region discussion following that theorem. The authors explicitly state that their corollaries recover precisely the Type I and Type II ranges used under Selberg's eigenvalue conjecture and invoke the calculation in the proof of [6, Theorem 2]. □

We also use Merikoski's Fundamental Proposition II with Selberg parameter zero. In the present notation it evaluates

$$\sum_{u\le x^{\xi-\eta}} \lambda_u S(\mathcal{A}(P)_u, x^{\gamma-\eta}) \tag{2}$$

by its model, for divisor-bounded λ_u, where

$$\sigma = \frac{2-\alpha}{3}, \qquad \gamma = \sigma - \alpha + 1 = \frac{5-4\alpha}{3}, \qquad \xi = \frac{3}{2} - \alpha. \tag{3}$$

This follows from Proposition 2.1 by the proof of [6, Proposition 6].

3. The recursive switching

Fix $7/6 < \alpha < 5/4$ and put

$$a = \alpha - 1, \qquad \tau = \frac{\xi}{2}.$$

Then $0 < \gamma < \tau < a < \sigma$. We suppress the harmless interior margins in the displayed limiting formulae, but record one compatible choice here. If $\nu > 0$ is the parameter in Corollary 7.2 of [2], put

$$\begin{aligned} &a_\nu = a + 3\nu, \quad \sigma_\nu = \sigma - 2\nu, \\ &\gamma_\nu = \gamma - 4\nu, \quad \xi_\nu = \xi - 3\nu, \qquad \tau_\nu = \xi_\nu/2. \end{aligned} \tag{4}$$

The available Type II interval is $[a + 2\nu, \sigma - 4\nu/3]$, so $[a_\nu, \sigma_\nu]$ lies strictly inside it. The proof of the fundamental proposition works uniformly for $u \le x^{\xi_\nu}$ and sieve level x^{γ_ν}: its Type I part has

$$x^{a+2\nu} x^{\xi_\nu} = x^{1/2-\nu},$$

while the first Type II prefix in its complementary part is smaller than

$$x^{a+2\nu+\gamma_\nu} = x^{\sigma-2\nu} < x^{\sigma-4\nu/3}.$$

Thus the two published corollaries give exactly this margin version, not merely the limiting endpoint. On the compact range $\alpha < 1.22$ used by the certificate, all inequalities $0 < \gamma_\nu < \tau_\nu < a < \sigma$ hold once ν is sufficiently small. We use these shifted parameters in the analytic sums and let $\nu \to 0^+$ only after obtaining the uniform estimates.

3.1. **Uniform localization and prefix sieving.** We record two standard uniformity facts because the recursive decomposition uses them in combinations not written explicitly in the earlier papers. The first is the finite Mellin–Perron device used below.

Lemma 3.1 (Finite cross-condition localization)**.** *Fix $R, C, J \ge 1$. Consider a dyadically localized bilinear sum with $MN \asymp x^\alpha$, whose two coefficients are bounded by τ_R, and impose at most J strict conditions*

$$U(\boldsymbol{m}) < V(\boldsymbol{n}).$$

Here U and V are positive integer-valued functions of the variables assigned to the M- and N-coefficients, respectively, and $U, V \le x^C$. A fixed smooth function of MN/P and a factor $\log(MN)$ may also be present. Then the sum is a multiple Mellin–Perron integral of ordinary Type II sums whose coefficients are bounded by $O_{C,J}(\tau_R)$. The total L^1 mass of the kernels is $O((\log x)^{O_{C,J}(1)})$, and the truncation error is $O_A(x^{-A})$ for any fixed $A > 0$, after increasing the truncation height. Consequently any fixed power-saving Type II estimate that is uniform for divisor-bounded coefficients survives all these localizations.

Proof. Since U, V are integers, the strict comparison is equivalent to $U < V - 1/2$. Perron's formula with $c = 1/\log x$ and height T gives

$$\mathbf{1}_{U<V} = \frac{1}{2\pi i}\int_{c-iT}^{c+iT}\left(\frac{V-1/2}{U}\right)^s \frac{ds}{s} + O\left(\min\left\{1, \frac{1}{T\,|\log((V-1/2)/U)|}\right\}\right).$$

The replacement of V by $V-1/2$ only excludes the equality case. It does not change the underlying representations or their divisor bounds. The truncation estimate in the displayed Perron formula remains an analytic input and is not part of the Lean audit.

For a cross-condition $U(\boldsymbol{m}) < V(\boldsymbol{n})$, attach U^{-s} to the M-coefficient and $(V-1/2)^s$ to the N-coefficient. If the smaller variable belongs to the N side, attach U^{-s} there and $(V-1/2)^s$ to M; the designation of the squarefree Type-II family is not changed. A product cutoff has the same form: if $A_M A_N \le Q$, with A_M and A_N formed from variables on the two sides, write it as

$$A_M < \lfloor Q/A_N \rfloor + 1.$$

Thus $U = A_M$ is assigned to the M side and $V = \lfloor Q/A_N \rfloor + 1$ to the N side; the complementary inequality is handled by subtracting this class. This also covers any exact priority cutoff not already fixed by the exponent cell. If U or V depends on a particular prime-factor representation rather than only on the collected integer, attach the phase before collecting that coefficient. The number of prime slots is fixed and the vertical factors are uniformly bounded, so summing those representations preserves a fixed divisor bound.

Here is the specialization used by the recursive branches.

- For the A_3 Type-II branch, choose the first eligible subset I and put $N = \prod_{i\in I} q_i$ and $M = (\prod_{i\notin I} q_i)ds$. Every ordering comparison split across the two families uses the smaller prime as U and the larger as V; hence the corresponding U^{-s} and $(V-1/2)^s$ are placed on the coefficient that contains that prime. If $q_3 \in N$, the roughness condition is $U = P^+(d) < V = q_3$ for $d > 1$, with U^{-s} on the M side and $(q_3 - 1/2)^s$ on the N side; $d = 1$ is automatic. If $q_3 \notin N$, the condition is internal to M. Earlier-subset priority conditions are fixed on cells away from their boundaries; if retained exactly, their product cutoffs use the preceding $A_M A_N \le Q$ rule.
- The A_3 lower-sieve branch is a Type-I prefix problem, not an application of the present Type-II localization lemma. All three primes belong to the prefix $u = q_1q_2q_3$; their ordering is internal, and the variable roughness cutoff q_3 is encoded in the prefix-dependent Rosser weight of Proposition 3.2.
- The one-prime tail base has prefix $u = q_1$ and the fixed cutoff x^{γ_ν}. Its cell order $\tau_\nu < \beta_1 < a$ is fixed before the Fundamental Proposition is applied, so it introduces no cross-condition between two coefficient families.
- For the direct pair Type-II upper bound, put $N = q_1q_2$ and place the sifted cofactor and any Mobius variable on M. The ordering $q_2 < q_1$ is internal to N. The only variable roughness comparison is $P^+(d) < q_2$ for $d > 1$: take $U = P^+(d)$ on M and $V = q_2$ on N; again $d = 1$ is automatic.
- The universal pair bound U_{LS} is again Type I: the prefix is $u = q_1q_2$, and the least prefix prime q_2 is the cutoff in the prefix-dependent Rosser weight. No Type-II Perron separation is used in this branch.
- In the recursive-base option, the base has prefix $u = q_1q_2$ and the fixed cutoff x^{γ_ν}; it has no new cross-condition. Its subtracted three-prime children

reuse, branch for branch, the A_3 Type-II or lower-sieve assignments above. The priority decision between the direct and recursive upper candidates is fixed by the proof-adverse exponent box and is not a Perron condition.

For each priority class the number J of genuine cross-comparisons is fixed, independent of x. On $\Re s = c$, the real powers U^{-c} and $(V-1/2)^c$ are $O_C(1)$, while the imaginary powers have modulus one. Each Perron kernel has L^1 mass $O(\log T)$. Moreover $|V - 1/2 - U| \geq 1/2$ and $U, V \leq x^C$, so $|\log((V-1/2)/U)| \gg_C x^{-C}$. If the unreduced finite decomposition has at most x^B tuples, choose $T = x^{A'}$ with $A' > A + B + C$. Summing the pointwise truncation errors is then $O_A(x^{-A})$. Iterating at most J comparisons gives total kernel mass $O((\log x)^J)$, which is absorbed by the fixed power saving of Corollary 7.2 after decreasing its exponent.

For a smooth compactly supported W, Mellin inversion separates $W(MN/P)$; rapid decay of its Mellin transform gives the same arbitrary power truncation error. Finally $\log(MN) = \log M + \log N$, with the logarithms handled by partial summation. All vertical factors have size $O_C(1)$, so the coefficients remain uniformly divisor bounded. The fixed polylogarithmic kernel mass is absorbed after decreasing the power-saving exponent. This proves the lemma. Merikoski's use of Perron in the proof of Fundamental Proposition II and his Mellin removal of the smooth product weights provide the qualitative templates [6, Sections 2.3 and 3.4]. The explicit truncated formula, its aggregate error, and every branch assignment above remain analytic steps not checked in Lean. □

Proposition 3.2 (Prefix-uniform linear sieve)**.** *Let a dyadically localized ordered prefix $\boldsymbol{q} = (q_1, \ldots, q_j)$ consist of at most three primes. Put $u = q_1 \cdots q_j$ and suppose that the localized cell gives the uniform upper bound $u \leq x^{s_0}$, with $1/2 - s_0 - \nu > 0$. All prefix primes are at least the sieve cutoff $z = z(\boldsymbol{q})$, with $z(\boldsymbol{q}) \asymp x^{\zeta}$; in the variable cutoff branches, $z(\boldsymbol{q}) = q_j$ is the least prefix prime. The Type I estimate in Proposition 2.1, summed over any fixed finite collection of such cells with divisor-bounded prefix weights, permits the standard dimension-one upper and lower linear sieve with remainder level*

$$D = x^{1/2 - s_0 - \nu}.$$

After removal of the common Euler factor, the corresponding sieve parameter is $s = (1/2 - s_0 - \nu)/\zeta$ and the main-term multipliers are $e^{-\gamma_E} F(s)$ and $e^{-\gamma_E} f(s)$. The lower multiplier is understood to be zero for $s \leq 2$. The aggregate remainder is $O(x^{1-\delta})$ for some $\delta = \delta(\nu) > 0$.

Proof. For each localized prefix, insert the upper or lower Rosser weights at level $D = x^{1/2-s_0-\nu}$. The per-prefix inequalities are the dimension-one formulae (12.12)–(12.14) of [3, Chapter 12, p. 236]. Summing these inequalities first and then collecting $e = ud$ gives a coefficient $\Lambda_e^{\pm}$ depending only on the total modulus. The detailed argument in Appendix A proves

$$|\Lambda_e^{\pm}| \ll (\log x)^B \tau_K(e), \qquad ud \leq x^{1/2-\nu},$$

so the absolute discrepancy sum in [2, Corollary 7.1, p. 22], with $a = h = 1$, leaves a fixed power saving after all prefix cells are summed.

The same appendix treats the two uses separately. For the A_3 lower branch it proves that the three-prime level does not degenerate at $\alpha = 7/6$, and that $s \leq 2$ contributes zero. For the two-prime upper branch it obtains U_{LS}; when $0 < s < 1$, the cutoff is lowered to $z_* = D^{1-\omega}$ before the Chapter 12 upper sieve is applied. Merikoski's Lemma 7 then converts the summed model terms to (12) and (13). This proves the proposition subject to the external analytic inputs stated in Remark 3.3. □

Remark 3.3 (Limitation of the prefix-sieve transfer). Proposition 3.2 is an analytic input to the candidate theorem, not a Lean theorem. Its decisive step is that Corollary 7.1 may be applied after the prefix-dependent Rosser weights have been collected into $\Lambda_e^{\pm}$ and bounded as in (27). If that collected coefficient does not satisfy the divisor-bounded hypothesis uniformly, if the prefix-dependent inner weights cannot first be separated to produce a common discrepancy $r(e)$, or if the Chapter 12 fundamental lemma is not uniform for the least-prefix cutoff, then the A_3 lower-sieve and U_{LS} branches are not justified and the analytic argument does not close.

The decomposition preceding the old linear-sieve term is

$$S(\mathcal{A}(P), 2\sqrt{P}) \leq S(\mathcal{A}(P), x^a) - \sum_{x^a < q < x^\sigma} S(\mathcal{A}(P)_q, q). \tag{5}$$

The negative sum in (5) is retained unchanged; its model contribution is the term F_6 bounded in Section 5. We improve only the first term. The first Buchstab split is

$$\begin{aligned} S(\mathcal{A}(P), x^a) &= S(\mathcal{A}(P), x^\gamma) - \sum_{x^\gamma < q_1 < x^\tau} S(\mathcal{A}(P)_{q_1}, q_1) - R_1, \\ R_1 &:= \sum_{x^\tau < q_1 < x^a} S(\mathcal{A}(P)_{q_1}, q_1). \end{aligned} \tag{6}$$

Applying Buchstab twice more to the retained sum yields

$$S(\mathcal{A}(P), x^a) = A_0 - A_1 + A_2 - A_3 - R_1, \tag{7}$$

where

$$\begin{aligned} A_0 &= S(\mathcal{A}(P), x^\gamma), \\ A_1 &= \sum_{x^\gamma < q_1 < x^\tau} S(\mathcal{A}(P)_{q_1}, x^\gamma), \\ A_2 &= \sum_{x^\gamma < q_2 < q_1 < x^\tau} S(\mathcal{A}(P)_{q_1 q_2}, x^\gamma), \\ A_3 &= \sum_{x^\gamma < q_3 < q_2 < q_1 < x^\tau} S(\mathcal{A}(P)_{q_1 q_2 q_3}, q_3). \end{aligned}$$

The first three terms are covered by (2), since $q_1 \leq x^\tau$ and $q_1 q_2 \leq x^{2\tau} = x^\xi$.

For A_3, retain only tuples for which at least one of

$$\beta_1 + \beta_2, \quad \beta_1 + \beta_3, \quad \beta_2 + \beta_3, \quad \beta_1 + \beta_2 + \beta_3 \tag{8}$$

lies in the shifted Type II interval $[a_\nu, \sigma_\nu]$. The corresponding product of distinct primes is squarefree and may be chosen as the Type II variable. Fix a sufficiently fine exponent mesh and discard cells meeting a Type II endpoint or a priority boundary. This is one-sided in the required direction, since we seek a lower bound for A_3; the model loss tends to zero with the mesh by dominated convergence. On every remaining cell, Lemma 3.1 removes the finitely many ordering and roughness cross-conditions and leaves divisor-bounded coefficients. This Type II part gives one lower bound for A_3. Unlike the earlier three-step argument, we also lower-bound complementary triples by the dimension-one lower linear sieve and take the larger available weight. Since A_3 occurs with a negative sign, every such lower bound may be subtracted in (7).

Lemma 3.4. *After restoring fixed interior margins, the retained part of A_3 has the same asymptotic formula for $\mathcal{A}(P)$ and $\mathcal{B}(P)$, with a power-saving error on each smooth P-block.*

Proof. Order the four subsets in (8) and assign each good prime triple to the first subset whose exponent sum lies in the fixed interior of $[a_\nu, \sigma_\nu]$. For every earlier subset, nonmembership is the disjoint union of the alternatives "below the lower endpoint" and "above the upper endpoint". Expanding these alternatives produces only finitely many priority classes (at most $1+2+2^2+2^3$ before empty classes are removed). Smoothly localize the three prime variables. For an assigned subset I, put $N = \prod_{i\in I} q_i$ and put the complementary prime factors and the sifted cofactor into M. Then $MN \asymp x^\alpha$,

$$x^{a_\nu} \le N \le x^{\sigma_\nu}, \qquad M \ge N,$$

and N is squarefree. The coefficient of N has only divisor-boundedly many prime representations. The coefficient of M is also divisor bounded, since it records at most the complementary prime factors and one cofactor.

Here are the coefficient and roughness details. From the definition of a sifted sum,

$$S(\mathcal{A}(P)_{q_1q_2q_3}, q_3) = \sum_{(r,P(q_3))=1} |\mathcal{A}_{q_1q_2q_3r}|\psi_P(q_1q_2q_3r)\log(q_1q_2q_3r).$$

Insert the finite identity

$$\mathbf{1}_{(r,P(q_3))=1} = \sum_{d|r,\, d|P(q_3)} \mu(d)$$

and write $r = ds$. If $N = \prod_{i\in I} q_i$ is the selected subset, put $M = (\prod_{i\notin I} q_i)ds$. After dyadic subdivision, the resulting sum has exactly the bilinear form of Proposition 2.1. The N-coefficient is supported on squarefree integers: the selected q_i are distinct. More quantitatively, if $t = |I|$, then before the separation of cross-conditions the numbers of possible factorizations obey the uniform bounds

$$\#\{N = \textstyle\prod_{i\in I} q_i\} \le \tau_t(N), \qquad \textstyle\sum_{M=(\prod_{i\notin I} q_i)ds} |\mu(d)| \le \tau_{5-t}(M).$$

The harmless conventions $\tau_0(1) = 1$ and $\tau_j \le \tau_5$ may be used. Thus both coefficients are bounded by a fixed divisor function. Repeated prime factors shared by M and N do not affect this conclusion: Corollary 7.2 requires squarefree support only for the N coefficient and imposes no coprimality condition on M, N.

There is one genuine cross-condition when q_3 belongs to the selected product N. For $d > 1$ the condition $d \mid P(q_3)$ is equivalently

$$\mu(d) \ne 0, \qquad P^+(d) < q_3;$$

the term $d = 1$ is automatic. Hence the only datum from the M side which must be compared with the N side is the single integer $P^+(d)$. A truncated Perron integral separates $P^+(d) < q_3$ into $(P^+(d))^{it}$ and q_3^{-it}. If $q_3 \notin N$, this condition is already internal to the M coefficient. This makes explicit that expanding the roughness condition does not create an unbounded number of cross variables.

It remains to separate conditions that involve variables on both sides. Prime ordering and the strict comparison $P^+(d) < q_3$ have exactly the integer form covered by Lemma 3.1; using $q_3 - 1/2$ in the latter comparison also removes any equality ambiguity in truncated Perron inversion. The selected-subset priority is already fixed on the interior exponent cell. The product cutoff and logarithm are handled by the Mellin and partial-summation separations in that lemma. Hence the two coefficient bounds above remain $O(\tau_5(M))$ and $O(\tau_5(N))$, while all kernel and dyadic losses are powers of $\log x$. These are absorbed by the fixed power saving in Proposition 2.1.

This is precisely the mechanism in the proof of Merikoski's Fundamental Proposition and in the evaluation of its three-prime term [6, Sections 2.3, 2.4.3, and 3.4]. Most importantly, the paper explicitly states there that *any combination* of q_1, q_2, q_3 lying in the Type II range may be used. The new term changes only the outer exponent boxes; every retained box lies in the unconditional interior supplied by Proposition 2.1. Summing the four disjoint priority classes proves the lemma. □

All conversions of model almost-prime sums below use exactly [6, Lemma 7]: its rough-number input is equation (2.4), its measure is $d\beta_1 \cdots d\beta_k/(\beta_1 \cdots \beta_{k-1}\beta_k^2)$, and the numerical Buchstab bounds used here are precisely equation (2.5). No stronger model-integral statement is assumed.

Let ω be the Buchstab function and set

$$u_+ = 0.5617, \qquad u_- = 0.5607.$$

The bounds

$$\omega(v) \le u_+ \quad (v \ge 4), \qquad \omega(v) \ge u_- \quad (v \ge 2.47) \tag{9}$$

are quoted in [6, (2.5)]. On the retained range $\alpha < 1.22$, an ordered three-prime child has

$$\frac{\alpha - \beta_1 - \beta_2 - \beta_3}{\beta_3} > \frac{3 - 2\alpha}{\alpha - 1} > 2.54.$$

The one- and two-prime fundamental-proposition bases have arguments at least 9 and 7.5, respectively. If $\beta_1 + \beta_2 \le \sigma$, the ordered Type II pair has argument at least $2(\alpha - \sigma)/\sigma \ge 6.4$. Thus every use of (9) is within its stated range.

We also use the standard dimension-one linear-sieve functions F and f [3, Chapter 12]. Define the rational lower envelope

$$\Phi(s) = \begin{cases} 0, & s < 2, \\ 4(s-2)/s^2, & 2 \le s \le 4, \\ 1/2, & s \ge 4. \end{cases} \tag{10}$$

For $2 \le s \le 4$ the identity $e^{-\gamma_E} f(s) = 2\log(s-1)/s$ and $\log(1+t) \ge 2t/(2+t)$ give

$$\Phi(s) \le e^{-\gamma_E} f(s). \tag{11}$$

At $s = 4$ the rational value is $1/2$, and the standard monotonicity of f gives the continuation in (10). Similarly, $e^{-\gamma_E} F(s) = 2/s$ for $1 \le s \le 3$ and is at most $2/3$ for $s \ge 3$.

For an ordered triple put

$$s_3 = \frac{1/2 - \beta_1 - \beta_2 - \beta_3}{\beta_3}, \qquad W_3 = \max\{u_- \mathbf{1}_{\text{good}}, \Phi(s_3)\},$$

where $\mathbf{1}_{\mathrm{good}}$ indicates that one of (8) lies in $[a,\sigma]$. Proposition 2.1 handles the first weight; Proposition 3.2 handles the second. Indeed, after a fixed prefix of total exponent s_0, the available sieve level is $x^{1/2-s_0-\nu}$, and combining its Rosser weight with at most three prime indicators still gives a fixed divisor-bounded coefficient. Hence

$$A_3^{\mathrm{low}}(\alpha) = \alpha \int_{\gamma<\beta_3<\beta_2<\beta_1<\tau} W_3 \,\frac{d\beta_1\, d\beta_2\, d\beta_3}{\beta_1\beta_2\beta_3^2} \tag{12}$$

is a valid limiting lower model for A_3.

It remains to retain part of R_1. For $\tau<\beta_1<a$, Buchstab gives

$$S(\mathcal{A}(P)_{q_1}, q_1) = S(\mathcal{A}(P)_{q_1}, x^{\gamma}) - \sum_{x^{\gamma}<q_2<q_1} S(\mathcal{A}(P)_{q_1q_2}, q_2).$$

Put $\delta_2 = 1/2-\beta_1-\beta_2$ and define the proof-adverse upper linear-sieve envelope

$$U_{\mathrm{LS}}(\beta_1,\beta_2) = \max\left\{\frac{2}{\delta_2}, \frac{2}{3\beta_2}\right\}.$$

Let $U_2(\beta_1,\beta_2)$ be the minimum of U_{LS} and the following available upper bounds:

$$\begin{aligned} &\frac{u_+}{\beta_2}, && \text{if } \beta_1+\beta_2\in[a,\sigma],\\ &\frac{u_+}{\gamma} - \int_{\gamma}^{\beta_2} W_3(\beta_1,\beta_2,\beta_3)\frac{d\beta_3}{\beta_3^2}, && \text{if } \beta_1+\beta_2\le\xi, \end{aligned}$$

where the second candidate is used only when it is nonnegative. The second line is one more Buchstab identity: Fundamental Proposition II evaluates its base and (12) lower-bounds its children. The proof-adverse linear-sieve option is uniform over the two-prime prefix by Proposition 3.2. Consequently

$$T(\alpha) = \alpha\int_{\tau}^{a}\left[\frac{u_-}{\gamma} - \int_{\gamma}^{\beta_1} U_2(\beta_1,\beta_2)\frac{d\beta_2}{\beta_2}\right]_+ \frac{d\beta_1}{\beta_1} \tag{13}$$

is a lower model for R_1.

3.2. **Complete branch-to-input ledger.** Put

$$\mathfrak{M}(P) := X_b \int_0^{\infty} \psi_P(u)\,\frac{du}{u}.$$

This is the common smooth block factor in [6, Lemma 7]. For a fixed $\nu>0$ and a fixed exponent mesh, let $H_{\nu}(\alpha)$ denote the expression below with the shifted parameters in (4) and the proof-adverse cell endpoints used by the certificate.

Proposition 3.5 (Analytic branch ledger)**.** *Uniformly for* $7/6<\alpha<1.22$*, the recursive decomposition uses exactly the one-sided estimates in the following table. Each row is valid with a power-saving remainder before the finite sums of dyadic and Perron pieces.*

term	*analytic input*	*proof direction and condition*
A_0	*Fundamental Proposition II*	*upper bound,* $u = 1$*, using* u_+/γ
A_1	*Fundamental Proposition II*	*lower bound on the common one-prime domain, using* u_-/γ
A_2	*Fundamental Proposition II*	*upper bound on a containing ordered two-prime domain, using* $u_+/(2\gamma)$ *after symmetry*
A_3	*Corollary 7.2*	*lower bound whenever a complete subset-product cell lies in* $[a_\nu, \sigma_\nu]$
A_3	*Proposition 3.2*	*lower linear-sieve bound on the same child; take the larger of the two available lower bounds*
R_1 *base*	*Fundamental Proposition II*	*lower bound because* $\beta_1 < a < \xi$
two-prime child	*Proposition 3.2*	*universal upper linear-sieve envelope* U_{LS}
two-prime child	*Corollary 7.2 or one more Buchstab split*	*take the smaller upper bound: direct Type II when* $\beta_1 + \beta_2 \in [a_\nu, \sigma_\nu]$*; otherwise base upper minus* A_3 *lower when* $\beta_1 + \beta_2 \le \xi_\nu$

Consequently, in the normalization above,

$$S(\mathcal{A}(P), x^a) \le \big(H_\nu(\alpha) + o(1)\big)\mathfrak{M}(P),$$

where the $o(1)$ *is uniform on compact shifted cells. Letting the fixed margin and mesh width tend to zero gives* $H(\alpha)$ *below.*

Proof. The signs $A_0 - A_1 + A_2 - A_3 - R_1$ determine the required directions. The first three rows follow from (2), since their prefix exponents are at most $0, \tau$, and $2\tau = \xi$, respectively. The two A_3 rows are Lemma 3.4 and Proposition 3.2; the maximum of two lower bounds for the same nonnegative child remains a lower bound.

For R_1, Buchstab's identity writes each node as a base minus its two-prime children. A base lower bound minus children upper bounds is therefore a node lower bound. Each child has the universal upper linear-sieve bound. On a complete Type II pair cell, Corollary 7.2 supplies the second upper bound. If the pair exponent is at most ξ_ν, one more Buchstab identity instead gives a base upper bound minus the already proved three-prime lower bound. Taking the minimum of valid upper bounds is valid. Lemma 3.1 preserves all these estimates after the finitely many cross-condition separations. Summing the rows proves the claim. □

Remark 3.6 (Exhaustive branch coverage). The supplementary branch-coverage audit checks the complete canonical grid using integer comparisons only. It visits 34 215 168 ordered A_3 boxes and 19 635 200 two-prime tail boxes. Every row of the ledger occurs. The proof-adverse minimum Buchstab arguments are, respectively,

$$5.206977117, \qquad 9.007506255, \qquad 8.002880298, \qquad 7.508771632$$

for the A_3 Type II lower branch, the R_1 base, the direct pair Type II upper branch, and the recursive-base upper branch. Thus the first two exceed 2.47 and the last two exceed 4, as required for the constants u_- and u_+. The audit terminates with `BRANCH_LEDGER_CERTIFIED=YES`.

Finally put $L = \log(\tau/\gamma)$. The complete recursive upper envelope is

$$H(\alpha) = \frac{\alpha}{\gamma}\left(u_+ - u_- L + \frac{u_+}{2}L^2\right) - A_3^{\mathrm{low}}(\alpha) - T(\alpha). \tag{14}$$

The original linear-sieve envelope is 4α, so the new saving is

$$\Delta = \int_{7/6}^{5/4} \max\{0, 4\alpha - H(\alpha)\}\, d\alpha. \tag{15}$$

4. Integer certificate for the switching integral

Proposition 4.1. *The integral in* (15) *satisfies*

$$\Delta \geq \frac{32303187971}{10^{12}} = 0.032303187971 > \frac{4}{125}.$$

Proof. We describe the exact certificate implemented in the accompanying file `scripts/certify_harman_recursive_tail.cpp`. It uses $N = 1200$ equal α-cells, $B = 300$ cells for (12), and $C = 160$ cells for the tail (13). All quantities are stored as integer multiples of 10^{-12}. Products use signed 128-bit intermediates; lower bounds are rounded down, upper bounds are rounded up, and every narrowing conversion is checked for overflow.

For an α-cell $[\alpha_i, \alpha_{i+1}]$, monotonicity gives the common domain

$$[\gamma(\alpha_i), \tau(\alpha_{i+1})]$$

and the containing domain

$$[\gamma(\alpha_{i+1}), \tau(\alpha_i)].$$

The first three terms in (14) are bounded, in proof-adverse directions, by

$$\begin{aligned} A_{0,i}^{+} &= u_+ \frac{\alpha_{i+1}}{\gamma(\alpha_{i+1})}, \\ A_{1,i}^{-} &= u_- \frac{\alpha_i}{\gamma(\alpha_i)} R_B^{-}, \\ A_{2,i}^{+} &= \frac{u_+}{2} \frac{\alpha_{i+1}}{\gamma(\alpha_{i+1})} (R_B^{+})^2, \end{aligned}$$

where R_B^- is the right-endpoint lower Riemann sum for $\int d\beta/\beta$ over the common domain, and R_B^+ is the left-endpoint upper sum over the containing domain.

For the three-prime lower term, each ordered β_1, β_2 box produces a finite family of rational β_3 intervals. Some are uniformly Type II throughout the entire α, β_1, β_2 cell. The others come from the 21 rational lower-sieve thresholds between 2.01 and 4. Overlapping intervals take the largest certified lower weight. The β_3 integral is evaluated exactly by

$$\int_r^s \frac{d\beta_3}{\beta_3^2} = \frac{1}{r} - \frac{1}{s},$$

and $1/(\beta_1\beta_2)$ is replaced by its value at the upper box endpoints. Thus the resulting $A_{3,i}^-$ is a lower bound.

For the recursive tail, a β_1 cell lies in the common intersection $[\tau(\alpha_i), a(\alpha_i)]$. Its child β_2 cells cover a containing domain from $\gamma(\alpha_{i+1})$ to the upper endpoint of the β_1 cell. Each child upper bound is the minimum of the proof-adverse linear-sieve bound, a uniformly available Type II bound, and the recursive base-minus-children bound when the two-prime prefix is at most ξ. The same-exponent-cell region is subtracted in full. This deliberately overcounts both distinct primes in one cell and the harmless region $\beta_2 \geq \beta_1$; it cannot make the tail lower bound too large.

It follows pointwise on the cell that

$$\max\{0, 4\alpha - H(\alpha)\} \geq \max\{0, 4\alpha_i - (A^+_{0,i} - A^-_{1,i} + A^+_{2,i} - A^-_{3,i} - T^-_i)\}.$$

Only $\alpha < 1.22$ is retained; zero is used elsewhere.

The exact integer output is

```
scale=1000000000000
alpha_steps=1200 beta_steps=300 tail_beta_steps=160
positive_alpha_cells=727
tail_lower_integral=0.018330264130
a3_lower_integral=0.044459636636
saving_lower_units=32303187971
saving_lower=0.032303187971
target=0.032000000000
CERTIFIED=YES
```

The last comparison is the integer inequality $32303187971 > 32000000000$.

The module `HarmanRecursiveCertificate.lean` is a third Lean implementation of the complete fixed-point algorithm. Lean evaluates the canonical grid by `native_decide` in four 192-cell blocks and one zero block. The respective unscaled saving sums are

$$154113574025565, \quad 151615832402575, \quad 117510700363272, \quad 41925800002266, \quad 0.$$

Their sum is 465165906793678; downward division by $12N = 14400$ is 32303187971. Small projection modules connect every expensive block computation to its exact natural-number value. Thus the large block equalities use Lean's native evaluator; after those equalities are available, `HarmanRecursiveCertificateCanonical.lean` checks the final natural-number sum, division, rational normalization, and strict endpoint inequalities by kernel elaboration without re-expanding the blocks.

As an implementation-level cross-check, the accompanying Python audit program independently reconstructs every cell using Python arbitrary-precision integers. It does not call or parse the C++ executable and uses an independent sweep-line algorithm for weighted interval maxima. On the canonical grid it reports the same totals. With the optional `-dump-cells` flag, all 1200 canonical cell records agree byte for byte; both dumps have SHA-256

`6e1901ca1dc8b2c32ae6bcc3759a42ea`
`9a3dab8932870eeb3f6e2d42f0ac3175`.

Thus the Lean, fixed-width, and arbitrary-precision implementations all recompute and certify the displayed lower bound independently, with the Lean block computations using `native_decide` as just stated. □

Remark 4.2. The same source compiled with GCC's undefined-behavior sanitizer gives the same result. The finer grids $1800 \times 400 \times 220$ and $2400 \times 500 \times 280$ give the respective lower bounds 0.033508163100 and 0.034213110567. These values support convergence but are not used in the theorem.

To pass from the analytic margin statement to the certified limiting integral, use the explicit parameters in (4) and denote the resulting region by G_ν. We keep only $\alpha < 1.22$, as does the certificate. On this compact range, γ is bounded away from zero and all integrands above have a common integrable bound. The Type II, lower-sieve threshold, ordering, and cell boundaries lie in a finite union of affine hyperplanes, a null set. Their fixed-margin indicators converge pointwise away from this set. Dominated convergence therefore shows that the full switched envelope with margins converges to (15) as $\nu \to 0^+$. The exact endpoint margin in (19) is $2994511739/9000000000000 > 0$; choose a fixed positive ν and a sufficiently fine fixed priority mesh for which the total integral loss is less than half this margin. After that choice, the power-saving and $o(1)$ errors are absorbed by taking x sufficiently large.

5. An independent lower bound for the old subtractive term

The original decomposition retains

$$F_6 = \int_{7/6}^{5/4} \alpha \int_{\alpha-1}^{\sigma} \omega\left(\frac{\alpha-\beta}{\beta}\right) \frac{d\beta}{\beta^2}\, d\alpha. \tag{16}$$

We need a stronger bound than the conservative 0.035631 printed in the original paper, but it can be obtained without numerical Buchstab integration.

Lemma 5.1. *One has*

$$F_6 > \frac{149403}{2500000} = 0.0597612.$$

Proof. Set $v = \alpha/\beta - 1$. The inner integral in (16) becomes

$$\int_{(4\alpha-2)/(2-\alpha)}^{1/(\alpha-1)} \omega(v)\, dv.$$

Throughout $7/6 \le \alpha \le 5/4$, this interval lies in $[3.2, 6]$. Thus the lower bound in (9) applies and

$$F_6 \ge 0.5607 \int_{7/6}^{5/4} w(\alpha)\, d\alpha, \qquad w(\alpha) = \frac{1}{\alpha-1} + 4 - \frac{6}{2-\alpha}.$$

Moreover,

$$w''(\alpha) = \frac{2}{(\alpha-1)^3} - \frac{12}{(2-\alpha)^3} > 0,$$

since the first term is at least 128 and the second at most $768/27$. Therefore the composite midpoint sum is a lower bound. Ten equal panels give, by exact rational arithmetic,

$$\begin{aligned} 0.5607 \int_{7/6}^{5/4} w(\alpha)\, d\alpha &\ge \frac{5607}{10000} \frac{1}{120} \sum_{j=0}^{9} w\left(\frac{7}{6} + \frac{2j+1}{240}\right) \\ &= 0.0597612998251544\ldots > 0.0597612. \end{aligned}$$

□

6. The endpoint budget and proof of the theorem

Merikoski's published strict upper bounds for the other terms are

$$F_1 < 0.0287, \qquad F_2 < 0.08622, \qquad F_3 < 0.03107,$$
$$F_4 < 0.00011, \qquad F_5 = \frac{29}{72}.$$

Combining these with Lemma 5.1 and the complete lower bound in Proposition 4.1, the normalized contribution through exponent $5/4$ is strictly smaller than

$$\begin{aligned} C &= \frac{1}{6} + \frac{287}{10000} + \frac{8622}{100000} + \frac{3107}{100000} + \frac{11}{100000} + \frac{29}{72} \\ &\quad - \frac{149403}{2500000} - \frac{32303187971}{10^{12}} \\ &= \frac{5611320508261}{9000000000000}. \end{aligned} \tag{17}$$

Above $5/4$, the unchanged dimension-one linear sieve contributes through a block exponent ϖ

$$4\int_{5/4}^{\varpi} \alpha \, d\alpha = 2\left(\varpi^2 - \left(\frac{5}{4}\right)^2\right). \tag{18}$$

Choose

$$\varpi_* := \frac{13231}{10000} = 1.3231.$$

Exact rational arithmetic gives

$$C + 2\left(\varpi_*^2 - \left(\frac{5}{4}\right)^2\right) = \frac{8997005488261}{9000000000000} < 1, \tag{19}$$

$$1 - \frac{8997005488261}{9000000000000} = \frac{2994511739}{9000000000000}. \tag{20}$$

Let P_x denote the greatest prime factor among the values n^2+1 selected by the smooth index block. The Chebyshev–Hooley identity used in [6] is

$$\sum_{x<p\le P_x} |\mathcal{A}_p| \log p = X_b \log x + O(x). \tag{21}$$

If $P_x \le x^{\varpi_*}$, summing the smooth modulus blocks and applying (17), (18), and the interior-margin discussion gives a strict upper bound smaller than $X_b \log x$, contradicting (21) for all sufficiently large x. Hence every sufficiently large index block $[x, 2x]$ contains an n such that

$$P^+(n^2+1) > x^{1.3231}. \tag{22}$$

Finally put $a = 1323/1000 = 1.323$ and $b = 13231/10000 = 1.3231$. Then $b - a = 1/10000$ and $a/(b-a) = 13230$. For $x > 2^{13230}$ and $n \le 2x$,

$$n^a \le (2x)^a = 2^a x^a < x^{b-a} x^a = x^b.$$

Together with (22), this proves Theorem 1.1.

7. Formal verification and scope

The focused Lean project contains the finite weighted Buchstab identity, its disjoint first-bad-index iteration, the sign pattern $A_0 - A_1 + A_2 - A_3$, the recursive node upper/lower rules, and the subtractive-tail inequality used in Section 3. Its rational-envelope module proves the elementary lower envelope used in the finite certificate. The recursive-certificate module and its four block modules independently recompute the exact outward-rounded certificate; the value and canonical modules aggregate the block totals while keeping peak elaboration memory bounded. The F_6 module proves positivity of the displayed second derivative, convexity, the ten-panel midpoint integral lower bound, and the exact rational comparison with 149403/2500000; applying this width certificate to F_6 still uses the quoted Buchstab lower bound. The endpoint module checks the exact rational comparisons, including $1323/1000 < 13231/10000$ and the strict normalized budget. The real-power module proves the final implication $x > 2^{13230}$ and $0 \le n \le 2x$ imply $n^{1323/1000} < x^{13231/10000}$. The focused project builds with Lean 4 and Mathlib v4.22.0.

Small audit modules check only affine identities for the ν margins, a half-integer cutoff, an abstract count of representations, one-sided box geometry, and algebraic normalization. The normalized expressions are α/γ, $t+1$, Φ, and U_{LS}. These modules construct neither the actual Perron localization nor Rosser coefficients varying with the prefix, and they do not prove any of the six analytic transfer gates as a whole.

The formal development treats Grimmelt–Merikoski Corollaries 7.1 and 7.2, their application to the actual localized coefficients in Lemma 3.4, the truncated Perron/Mellin transfer for the real sifted sums, and the standard upper and lower dimension-one linear-sieve theorem as analytic inputs. The finite Buchstab decomposition, the integer computation, its primitive outward-rounding rules, the elementary F_6 convex midpoint certificate, the exact rational endpoint arithmetic, and the real-power dyadic conversion are checked in Lean. The expensive $1200 \times 300 \times 160$ block equalities are discharged by `native_decide`; the canonical sum, division, and endpoint comparisons are then checked without native recomputation. Thus the native compiler is a trust boundary only for block values independently cross-checked by C++ and Python; downstream arithmetic and the finite lemmas on signs, widths, clipping, and endpoint directions are kernel checked. None of these facts asserts that the integer algorithm equals the analytic $H(\alpha)$ integral. No floating-point or proprietary computer-algebra calculation is an input to the finite certificate.

Remark 7.1 (AI-assisted proof development). The author discloses that GPT-5.6 Sol was used as a reasoning and coding assistant in developing the core recursive Buchstab switching argument, the proof-direction and branch ledger, the Lean certificate implementation, the exact-integer audit scripts, and the manuscript exposition. The author reviewed the final mathematical statements and takes responsibility for the submitted content. GPT-5.6 Sol is an AI-assisted development tool, not an author or an independent verifier. The Lean project checks only the finite combinatorial, integer, rounding, elementary F_6, rational endpoint, and real-power conversion components; the cited analytic Type-I/II, linear-sieve, and Mellin–Perron transfers remain external inputs.

Remark 7.2 (Supplementary material). The accompanying source archive contains the Lean project and locked toolchain, the fixed-width C++ certificate `certify_harman_recursive_tail.cpp`, and the independent arbitrary-precision Python audit `audit_harman_recursive_tail.py`. The command line `1200 300 160` selects the canonical grid in both external implementations; each prints `saving_lower` = `0.032303187971` and `CERTIFIED=YES`. No precomputed binary data are needed. The immutable source archive is `https://doi.org/10.5281/zenodo.22140875`. The separate script `audit_harman_recursive_branch_coverage.py` exhaustively checks the analytic branch gates summarized in Remark 3.6. A supplied build script packages the paper, PDF, locked Lean project, audits, and analytic ledger into one checksummed review archive. The archive also includes a source-version ledger with theorem locations and SHA-256 hashes for the external PDFs; the external PDFs themselves are not redistributed. The public verification repository is `https://github.com/CGandGameEngineLearner/landau-fourth-problem-1.323`.

Remark 7.3 (Analytic review boundary). The finite cross-condition localization and prefix-uniform sieve are proved in Lemma 3.1 and Proposition 3.2, but they are not formalized in Lean. Together with the quoted Type I/II and linear-sieve theorems, these are the parts that require conventional analytic-number-theory refereeing.

Remark 7.4 (Distance from Landau's problem). Theorem 1.1 implies infinitely often a factorization

$$n^2 + 1 = pm, \qquad p > n^{1.323}, \qquad m < n^{0.677+o(1)},$$

but it neither forces $m = 1$ nor overcomes the parity barrier. It is a new greatest-prime-factor weak form of Landau's fourth problem, not a proof of the prime-values conjecture.

Appendix A. Detailed prefix-uniform linear-sieve argument

This appendix gives the full bookkeeping behind Proposition 3.2. It is separated from the main argument for readability; its analytic inputs and limitations are unchanged.

The sequence for one prefix.
Fix one smooth dyadic cell and a prefix $\boldsymbol{q} = (q_1, \dots, q_j)$, $j \le 3$, with product u. The nonnegative sequence to be sifted is

$$a_{\boldsymbol{q}}(m) := |\mathcal{A}_{um}|\, \psi_P(um) \log(um)\, w_{\boldsymbol{q}}(m),$$

where $w_{\boldsymbol{q}}$ denotes the fixed smooth cofactor localization; removing it by Mellin inversion gives the same formula with a fixed polylogarithmic loss. For squarefree $d \mid P(z(\boldsymbol{q}))$, put

$$|\mathcal{A}_{\boldsymbol{q},d}| := \sum_{d|m} a_{\boldsymbol{q}}(m).$$

The corresponding model divisibility sum is

$$|\mathcal{B}_{\boldsymbol{q},d}| := X_b \sum_{d|m} \frac{\rho(um)}{um} \psi_P(um) \log(um)\, w_{\boldsymbol{q}}(m),$$

and define the unsifted localized model mass by

$$X_{\boldsymbol{q}} := |\mathcal{B}_{\boldsymbol{q},1}|.$$

Write

$$r_{\boldsymbol{q}}(d) := |\mathcal{A}_{\boldsymbol{q},d}| - |\mathcal{B}_{\boldsymbol{q},d}|. \tag{23}$$

This is exactly the actual-minus-model discrepancy to which the Type-I input is applied; we do not replace it by a pointwise assertion $g(d) = \rho(d)/d$. Every prime factor of d is strictly below the least prefix prime q_j, while every prime factor of u is at least q_j; hence $(u, d) = 1$.

After the smooth product and logarithm have been separated as in [6, Sections 2.3 and 3.4], evaluate the weighted model sum over d by the same Euler-product calculation as [6, Lemma 8] and [3, Theorem 11.12]. This calculation furnishes a dimension-one local density $g(p)$ and a localized model mass $X_{\boldsymbol{q}}$. Since $p < z$ implies $p \nmid u$, none of these local factors is changed by the prefix, and

$$V_{\boldsymbol{q}}(z) := \prod_{\substack{p<z(\boldsymbol{q}) \\ p\nmid u}} (1 - g(p)) = \prod_{p<z(\boldsymbol{q})} (1 - g(p)).$$

This is the same $V(z)$ and common Euler normalization as in Merikoski's Lemma 8. Only $X_{\boldsymbol{q}}$ depends on the localized prefix. The dimension-one hypothesis on the Euler product is unchanged for primes below the cutoff.

For this individual prefix, insert the upper or lower Rosser weights $\lambda^{\pm}_{d,\boldsymbol{q}}$ of level D. The weights are supported on squarefree $d \mid P(z(\boldsymbol{q}))$, $d < D$, and satisfy $|\lambda^{\pm}_{d,\boldsymbol{q}}| \le 1$. Their defining inequalities give

$$S(\mathcal{A}(P)_u, z) \le \sum_{\substack{d|P(z) \\ d<D}} \lambda^{+}_{d,\boldsymbol{q}} |\mathcal{A}_{\boldsymbol{q},d}|, \qquad S(\mathcal{A}(P)_u, z) \ge \sum_{\substack{d|P(z) \\ d<D}} \lambda^{-}_{d,\boldsymbol{q}} |\mathcal{A}_{\boldsymbol{q},d}|.$$

Apply the model Euler-product evaluation to the corresponding sums with $|\mathcal{B}_{\boldsymbol{q},d}|$, and use (23) for the difference. Equations (12.12)–(12.14) on p. 236 of [3, Chapter 12] then give, with $s_{\boldsymbol{q}} = \log D / \log z(\boldsymbol{q})$,

$$S(\mathcal{A}(P)_u, z(\boldsymbol{q})) \le X_{\boldsymbol{q}} V_{\boldsymbol{q}}(z) \Big(F(s_{\boldsymbol{q}}) + O((\log D)^{-1/6}) \Big) + |R^{+}_{\boldsymbol{q}}|, \tag{24}$$

$$S(\mathcal{A}(P)_u, z(\boldsymbol{q})) \ge X_{\boldsymbol{q}} V_{\boldsymbol{q}}(z) \Big(f(s_{\boldsymbol{q}}) - O((\log D)^{-1/6}) \Big) - |R^{-}_{\boldsymbol{q}}|, \tag{25}$$

where

$$R^{\pm}_{\boldsymbol{q}} := \sum_{\substack{d|P(z(\boldsymbol{q})) \\ d<D}} \lambda^{\pm}_{d,\boldsymbol{q}} r_{\boldsymbol{q}}(d), \qquad |R^{\pm}_{\boldsymbol{q}}| \le \sum_{\substack{d|P(z(\boldsymbol{q})) \\ d<D}} |r_{\boldsymbol{q}}(d)|.$$

The functions F, f are the dimension-one linear-sieve functions in [3, Chapter 12, (12.1)–(12.2), p. 235]; in particular (24) is used directly for $s_{\boldsymbol{q}} \ge 1$, and (25) for $s_{\boldsymbol{q}} \ge 2$. When $s < 2$ the lower bound used here is the trivial bound zero, recorded by the convention $f(s) = 0$. These displayed inequalities are the precise use of the Iwaniec–Rosser sieve in this paper.

Summing the prefixes and the Type-I remainder.

Now multiply (24) or (25) by the nonnegative localized prefix weight $c(\boldsymbol{q})$ and sum over the ordered prefixes in the cell. By the proof-adverse choice of its upper endpoint,

$$u \le x^{s_0}, \qquad D = x^{1/2-s_0-\nu}, \qquad ud \le x^{1/2-\nu}.$$

After collecting equal total moduli $e = ud$, the aggregate remainder is

$$\mathcal{R}^{\pm} = \sum_{e \le x^{1/2-\nu}} \Lambda^{\pm}_e r(e) + O_A(x^{-A}), \qquad \Lambda^{\pm}_e := \sum_{\boldsymbol{q},d:\, ud=e} c(\boldsymbol{q}) \lambda^{\pm}_{d,\boldsymbol{q}}, \tag{26}$$

where $r(e)$ is the modulus-e discrepancy in [2, Corollary 7.1, p. 22] with $a = h = 1$; the $O_A(x^{-A})$ term is the fixed Mellin/partial-summation separation. This formula is asserted only after every prefix-dependent cofactor cutoff, smooth product weight, and ordering condition has been separated as in Lemma 3.1. The resulting vertical parameters are absorbed into $c(\boldsymbol{q})$, uniformly in their integration ranges. Without this separation, $r_{\boldsymbol{q}}(d)$ need not be a function of $e = ud$ alone and Corollary 7.1 could not be applied in the displayed form.

For fixed e, each of the at most three ordered prefix primes is a divisor of e, and $d = e/u$ is then determined. Thus, if $|c(\boldsymbol{q})| \ll \tau_{K_0}(u)$, the fixed number of ordered representations and $|\lambda^{\pm}_{d,\boldsymbol{q}}| \le 1$ give

$$|\Lambda^{\pm}_e| \ll (\log x)^B \tau_K(e) \tag{27}$$

with fixed B, K, independent of x. This is the essential point for the variable cutoff: the weights $\lambda^{\pm}_{d,\boldsymbol{q}}$ may depend on $\boldsymbol{q}$, because Corollary 7.1 is applied only after they have been collected into $\Lambda^{\pm}_e$.

Write $\theta = 7/64$ and choose the auxiliary $\varepsilon_{\mathrm{GM}}$ in Corollary 7.1 so that

$$\delta_0 := (1 - 2\theta)\nu - \varepsilon_{\mathrm{GM}}/4 > 0.$$

That corollary gives

$$\sum_{e \le x^{1/2-\nu}} |r(e)| \ll x^{1-\delta_0}.$$

For any $\kappa > 0$, $\tau_K(e) \ll_{K,\kappa} e^{\kappa}$, and here $e \le x^{1/2-\nu} < x^{1/2}$. Choosing $\kappa < 2\delta_0$ in (27), and then absorbing $(\log x)^B$ and the $O((\log x)^3)$ ordinary dyadic prefix ranges, yields

$$|\mathcal{R}^{\pm}| \ll (\log x)^B x^{\kappa/2} \sum_{e \le x^{1/2-\nu}} |r(e)| \ll x^{1-\delta} \tag{28}$$

for some fixed $\delta = \delta(\nu) > 0$. Thus the remainder is small only after the whole multiple prefix sum is formed; no per-prefix power saving is claimed. The summed model main term on a fixed smooth P-block is of order x times the corresponding fixed model integral, so (28) is absorbed as $x \to \infty$. On every retained cell D is a fixed positive power of x; hence the $(\log D)^{-1/6}$ terms in (24)–(25) are uniform $o(1)$ multiples of the summed model main term.

The three-prime lower-sieve branch.
Here $u = q_1 q_2 q_3$, $z(\boldsymbol{q}) = q_3$, and s_0 is the upper endpoint for $\beta_1 + \beta_2 + \beta_3$. For fixed interior margin ν, the pointwise parameter and the proof-adverse cell parameter are respectively

$$\begin{aligned} s_{3,\nu} &= \frac{1/2 - \beta_1 - \beta_2 - \beta_3 - \nu}{\beta_3}, \\ \widehat{s}_{3,\nu} &= \frac{1/2 - s_0 - \nu}{\beta_3} + O(1/\log x) \le s_{3,\nu} + O(1/\log x). \end{aligned}$$

Summing (25) gives

$$\sum_{q_1,q_2,q_3} c(\boldsymbol{q}) S(\mathcal{A}(P)_{q_1q_2q_3}, q_3) \ge \sum_{q_1,q_2,q_3} c(\boldsymbol{q}) X_{\boldsymbol{q}} V_{\boldsymbol{q}}(q_3)\{f(\widehat{s}_{3,\nu}) + o(1)\} - O(x^{1-\delta}).$$

After removal of the common Euler factor, replace $e^{-\gamma_E} f(\widehat{s}_{3,\nu})$ by the rational minorant $\Phi(\widehat{s}_{3,\nu})$. The use of the cell upper endpoint makes this a lower, not an upper, weight. If $\widehat{s}_{3,\nu} \le 2$, then $f = 0$ and the lower bound used by the certificate is zero. In fact its first positive threshold is 2.01, so a degenerate cell cannot receive a positive lower-sieve weight.

There is also no loss of Type-I level at the endpoint $\alpha = 7/6$. Since

$$\tau_\nu = \frac{3/2-\alpha-3\nu}{2},$$

the worst three-prime cell satisfies $s_0 < 3\tau_\nu$; at $\alpha = 7/6$ one has

$$3\tau_\nu = \frac{1}{2} - \frac{9}{2}\nu, \qquad \frac{1}{2} - s_0 - \nu > \frac{7}{2}\nu.$$

For $\alpha > 7/6$ the available margin is larger. Thus D remains a positive power of x in the three-prime branch. The displayed limiting weight $\Phi(s_3)$ in (12) is obtained only after first taking $x \to \infty$ at fixed ν and then letting $\nu \to 0^+$; cells crossing $s = 2, 3, 4$ are taken in the one-sided direction and their boundary loss tends to zero.

The two-prime child upper sieve.

Here $u = q_1q_2$, $z(\boldsymbol{q}) = q_2$, and s_0 is the upper endpoint for $\beta_1+\beta_2$. On every retained box the proof-adverse mesh has a positive gap $1/2 - s_0$; since the mesh is finite, choose one ν smaller than the minimum of these positive gaps. Then

$$s_{2,\nu} = \frac{1/2-\beta_1-\beta_2-\nu}{\beta_2},$$
$$\widehat{s}_{2,\nu} = \frac{1/2-s_0-\nu}{\beta_2} + O(1/\log x) \le s_{2,\nu} + O(1/\log x),$$

and $\widehat{s}_{2,\nu} > 0$ for large x. On boxes with $\widehat{s}_{2,\nu} \ge 1$, summing (24) gives

$$\sum_{q_1,q_2} c(\boldsymbol{q}) S(\mathcal{A}(P)_{q_1q_2}, q_2) \le \sum_{q_1,q_2} c(\boldsymbol{q}) X_{\boldsymbol{q}} V_{\boldsymbol{q}}(q_2)\{F(\widehat{s}_{2,\nu}) + o(1)\} + O(x^{1-\delta}).$$

For $1 \le \widehat{s}_{2,\nu} \le 3$, equations (12.1)–(12.2) give $e^{-\gamma_E}F(\widehat{s}_{2,\nu}) = 2/\widehat{s}_{2,\nu}$; after the $d\beta_2/\beta_2$ measure this is $2/(1/2-s_0-\nu)$, which is at least the pointwise $2/(1/2-\beta_1-\beta_2-\nu)$. For $\widehat{s}_{2,\nu} \ge 3$, monotonicity gives the second upper coefficient $2/(3\beta_2)$.

It remains to cover $0 < \widehat{s}_{2,\nu} < 1$, where the Chapter 12 formula cannot be applied directly at $z = q_2$. Fix $\omega > 0$ and put $z_* := D^{1-\omega}$. For large x, $z_* < D < q_2$, and positivity gives

$$S(\mathcal{A}(P)_{q_1q_2}, q_2) \le S(\mathcal{A}(P)_{q_1q_2}, z_*).$$

The new sieve parameter is $s_* = 1/(1-\omega) > 1$, so (24) applies at z_*. Letting first $x \to \infty$ and then $\omega \to 0^+$ gives, after Euler normalization, the same coefficient $2/(1/2-s_0-\nu)$. Thus all positive-level boxes are covered by the proof-adverse fixed-margin envelope

$$U_{\mathrm{LS},\nu}(\beta_1,\beta_2) := \max\left\{\frac{2}{1/2-\beta_1-\beta_2-\nu}, \frac{2}{3\beta_2}\right\}.$$

Here decreasing the sieve parameter by using the upper endpoint s_0 can only enlarge the upper linear-sieve function; the containing beta endpoints used by the certificate enlarge the displayed envelope once more. If a box has no positive level after inserting the fixed margin, that box is not assigned a nontrivial sieve estimate; the certificate uses the containing upper branch or discards the associated tail saving. Letting $\nu \to 0^+$ after the uniform estimate gives exactly the displayed U_{LS} in (13). The transition at $s = 3$ is handled by taking the larger of the two upper expressions, which is the one-sided choice used by the integer certificate.

Model integrals and scope.
For $z(\boldsymbol{q}) \in [Z, 2Z]$, $Z = x^{\zeta}$, the exact parameter satisfies

$$\frac{\log D}{\log z(\boldsymbol{q})} = \frac{1/2 - s_0 - \nu}{\zeta} + O(1/\log x).$$

The model term is treated prefix by prefix before the remainders are collected. Merikoski's Lemma 7, with its hypotheses on smooth localization, ordered well-behaved prime regions, and fixed positive margins, converts the summed model terms into (12) and the U_{LS} contribution inside (13).

The proposition therefore uses two distinct external results: Chapter 12 of [3] for the per-prefix main terms (24)–(25), and Grimmelt–Merikoski Corollary 7.1 for the collected absolute remainder (28). Neither the prefix-dependent Rosser weights, this application of Corollary 7.1, nor the passage to the model integrals is formalized in Lean.

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